
WHITE PAPER · AI OPERATING GOVERNANCE

Average Accuracy Doesn't Tell You What You Could Lose

Measure the cases you accept, not every case scored

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EXECUTIVE SUMMARY

A model's average accuracy covers every case it scored. Before you commit more, track the acceptance rate, the error, and the money lost on the cases you actually accepted.

In June 2021 Zillow reported a 6.9 percent median error, the typical miss, across more than 104 million homes not listed for sale. In November it wrote down \$304 million on the homes it had bought, and its shareholder letter pointed to price forecasts that missed by more than modeled and to strained resale capacity. An average over every case a model scored need not describe the cases you accept, because the other side's yes and your own approval rule both choose them. Measure the accepted cases as their own group, and raise the cap on what you commit only while the numbers stay inside limits you set.

- 01** Write down who says yes to your model's estimates: your own approval rule, the other side, or both. The cases that get accepted can be a different group from the cases the model scored.
- 02** For the accepted cases, track three numbers: the acceptance rate against plan, the error (how far the estimate was from what happened), and the loss (money actually lost after margin and costs).
- 03** Cap what you commit when you sign, in cases or dollars. Raise the cap one step at a time, only after enough outcomes have arrived, and only while all three numbers stay inside limits you set before the increase.

What Zillow published and what it wrote down

In February 2021, Zillow began showing its home-value estimate, the Zestimate, as an initial cash offer.[1] On June 15 it announced a more accurate version. The national median error for off-market homes, meaning homes not listed for sale, fell to 6.9 percent across more than 104 million homes. That is the typical miss: half the estimates were off by less, and half by more. The release said Zillow began making live cash offers because of its "increasing confidence in Zestimate accuracy." [2]

On November 2, Zillow told shareholders it would shut down the program and cut about 25 percent of its workforce. In the third quarter it had bought 9,680 homes and

written down the value of the homes it held by \$304 million. A larger share of the owners who asked for an offer sold to Zillow than it had seen before, and the letter says that came “as a result of unintentionally purchasing homes at higher prices than our current estimates of future selling prices.” In plain words, its offers sat above what it later expected the homes to sell for, and more owners took them.[3]

Zillow gave other reasons too. It said it could not forecast home prices months ahead within the range it had modeled, and that buying so many homes strained its renovation and resale capacity. The letter does not split the loss among those causes. It also never mentions the 6.9 percent, which is about the Zestimate’s value for a home today. The write-down came from what Zillow expected homes to sell for months later.[3]

This paper starts from a question that applies to anyone who sizes a commitment from a model’s accuracy: accurate on which cases? The 6.9 percent covered more than 104 million homes. Zillow’s money sat in the homes whose owners said yes, and the letter says it converted only about 10 percent of the serious sellers who asked for an offer.[3] An average across the first group does not tell you how the estimates did on the second.

The people who say yes are not a random sample

Suppose a company buys used delivery vans. This example is made up. A model prices each van from its age, mileage, and model. It misses by \$1,000 high or \$1,000 low, in equal numbers. Across all the vans it prices, the average error is zero. The company offers the model’s price minus \$500 and plans to keep that \$500 as its margin on each van it buys.

Each owner knows things the model doesn’t, such as a weak transmission or a rusted frame, and has a fair idea what the van is worth. The owner of a van the model priced \$1,000 too high gets an offer \$500 above its worth, and says yes. The owner of a van priced \$1,000 too low gets an offer \$1,500 below its worth, and says no. The company ends up with the vans the model overpriced. It loses \$500 on each one instead of earning \$500, while the average error across everything the model priced is still zero.

Real markets are messier. Two studies look at the same problem, one by simulation and one with real purchases. Eirik Helgaker, Are Oust, and Arne Pollestad built three price models on 84,905 apartment sales in Oslo and simulated a home buyer making offers from them. When they assumed every offer was accepted, the buyer’s expected profit was 6.29 to 7.96 percent per apartment. When they assumed that owners, who know their homes better than a model does, accept overpriced offers

more readily than underpriced ones, expected profit fell to 0.19 to 1.21 percent.[4] That is a simulation with assumed owner behavior. It shows how large the effect could be under those assumptions, not what any real company lost. Economists call this adverse selection: the owners most eager to take an offer are often the ones who know it is too good.

Greg Buchak, Gregor Matvos, Tomasz Piskorski, and Amit Seru studied real purchases by iBuyers. These are companies that make fast cash offers from automated prices. They used property sale records from five U.S. markets between 2013 and 2018. iBuyers were least active where a standard price model missed the most. As they expanded into areas with bigger pricing errors, their profits fell, which the authors describe as consistent with adverse selection. Slow-selling homes are a second reason iBuyers hold back, they note.[5] Their reading, through their own model, is that conditions in late 2021 made the problem worse. They present that as an interpretation, not a finding about Zillow's books.[5]

So the average error across everything a model priced need not describe the cases that got accepted, and the gap can widen when a company moves into harder cases or when circumstances change.

Your own approval rule hides errors too

The other side is not the only one choosing. Your approval rule also decides which results you ever see.

EXHIBIT 1

Two choices decide which cases you end up with

The other side says yes

Owners who know more than the model tend to take the offers it overpriced, so accepted cases can be worse.

Your own rule says yes

You see results only for the cases you chose, so the model's record on the rest stays unknown.

Himabindu Lakkaraju and colleagues studied this in bail decisions. A judge releases some defendants and jails others, and the court learns whether a defendant skips a hearing or commits a new crime only if that person was released. Their data covered about 9,000 defendants and 18 judges in one district court. The cases with known outcomes are not a random sample of all defendants, and the authors warn that

checking a model only on cases with known outcomes “could potentially result in biased estimates” of how it performs.[6]

Business has the same problem. A lender sees how the loans it approved turn out, but not how the loans it refused would have turned out. A claims team sees how the claims it paid turn out, but not how the claims it denied would have turned out. A report that compares predictions with results is built from the cases you chose. It can make the model look better or worse than it is on the cases you turned away, and the report cannot tell you which. Your rule may even help. A screen that sends hard cases to people can lower the error on the cases you accept. Either way, the all-cases average will not show you.

Measure the cases you accept with three numbers

Start by writing down who says yes: your rule, the other side, or both. Then treat the accepted cases as their own group and track three numbers for it.

EXHIBIT 2

The three numbers, measured on the made-up vans

Acceptance rate

Owners of the overpriced half accept.
The rest decline: 50 percent.

Error on vans bought

\$1,000 too high on every van. Across
all vans priced, the average is \$0.

Loss on vans bought

\$500 lost on each van, instead of the \$500 margin the company planned to keep.

The first is the acceptance rate against your plan: the share of offers or applications that get accepted, compared with the share you expected. It is the only one of the three you can see before outcomes arrive. If it runs above plan, find out why the offers look good to the people accepting them. Zillow’s letter links its higher-than-anticipated conversion to the 9,680 purchases and the strain on its renovation and resale capacity.[3]

The second is error: how far the estimate was from what happened, such as the estimated resale price against the actual price.

The third is loss: the money actually lost on the accepted cases once margin, repairs, and holding costs are counted. In the van example the error was \$1,000 and the loss

was \$500, because the margin absorbed half of it. A model can miss by a lot and still make money if the margin is wide, or miss by a little and lose money if the margin is thin.

Where you can see how declined cases turned out, compare them with the accepted ones on the same model estimate. If the accepted cases did worse, the people saying yes may know something your model doesn't. This comparison is a suggested method, not a finding from the sources above.

Cap what you commit at signing, then widen the cap in steps

Set the limit at signing, because outcomes arrive after the commitments are made. Zillow's letter shows commitments running ahead of the results. It expected \$240 million to \$265 million in fourth-quarter losses "related primarily to inventory we have committed to purchase during Q4," and it had suspended signing new contracts only two weeks before the letter.[3]

Set a cap on the accepted group, in cases or dollars outstanding, small enough that a bad result is survivable. The leader who would bear the loss sets it, not the team that built the model. Before any increase, set a limit for each of the three. Raise the cap one step at a time, and only after enough outcomes arrive to judge error and loss. For homes, that means a full resale cycle. Lower the cap or stop when any of the three leaves its limit. A cap costs growth; weigh that against what a bad run could cost.

What to do next

Pick one workflow where a model's estimate sets a commitment that someone else can refuse: offers, quotes, credit limits, or automatically paid claims.

First, write down who says yes. Second, pull the accepted cases and, where you can see them, the declined ones. Compare the model's error and the money lost on each. Third, set the cap and the three limits, name the leader who owns them, and decide what evidence would let the cap rise.

If the accepted cases look like the whole group on error, and acceptance is on plan, the average is a fair guide and you can raise the cap in steps. If they don't, you know before the next increase.

Zillow's June release said how accurate the Zestimate was on 104 million homes. It did not say how accurate it was on the homes whose owners said yes. Ask for that number before you sign the next increase.

Notes and sources

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- [3] Zillow Group, "Q3 2021 Shareholder Letter," November 2, 2021, filed as Exhibit 99.3 to Form 8-K. <https://www.sec.gov/Archives/edgar/data/1617640/000161764021000085/exhibit993.htm>
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About the author

Marco Policani is an enterprise portfolio, PMO, and AI operating-governance leader. He builds portfolio governance systems where AI carries the analytical load and named humans own every decision — an approach documented across case studies, walkthroughs, and working governance guides on his portfolio site.

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